MA2508 Midterm Tutorial handnote

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Q1.

$$\frac{\partial z}{\partial x} = 2x f'(x^2-y^2)$$

$$\frac{\partial z}{\partial y} = 1-2y f'(x^2-y^2)$$

$$y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = 2xyf'(x^2-y^2) + x - 2xyf'(x^2-y^2) = x$$

Q2

$\vec{u}\cdot\vec{v}=0 \Longrightarrow \vec{u}\perp\vec{v}$ ①

$\vec{u}\times\vec{v}=0 \Longrightarrow \vec{u}\parallel\vec{v}$ ②

This is a contradiction if $\vec{u}\neq 0$ and $\vec{v}\neq 0$

Note: $\|\vec{u}\times\vec{v}\|^2 + (\vec{u}\cdot\vec{v})^2 = \|\vec{u}\|^2\|\vec{v}\|^2\sin^2\theta + \|\vec{u}\|^2\|\vec{v}\|^2\cos^2\theta$

$= \|\vec{u}\|^2\|\vec{v}\|^2$

if $\vec{u}=0$ or $\vec{v}=0$, then both ① and ② are true.

Q3

need to memorize this formula

$$\boxed{\kappa(t) = \frac{|\vec{r}\,'(t)\times\vec{r}\,''(t)|}{|\vec{r}\,'(t)|^3}}$$

we want $\kappa(0)$ when $t=0$

$\vec{r}\,'(t) = e^t(\cos t-\sin t,\ \sin t+\cos t,\ 1)$

$\vec{r}\,''(t) = e^t(-2\sin t,\ 2\cos t,\ 1)$

so $\begin{cases} \vec{r}\,'(0) = (1,1,1),\ |\vec{r}\,'(0)| = \sqrt{1^2+1^2+1^2} = \sqrt{3} \\ \vec{r}\,''(0) = (0,2,1) \end{cases}$

$\vec{r}\,'(0)\times\vec{r}\,''(0) = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 0 & 2 & 1 \end{vmatrix} = (1\times1-1\times2)i - (1\times1-0\times1)j + (1\times2-1\times0)k$

$= (-1,-1,2)$

$|\vec{r}\,'(0)\times\vec{r}\,''(0)| = \sqrt{(-1)^2+(-1)^2+2^2} = \sqrt{6}$

$$\kappa(0) = \frac{\sqrt{6}}{(\sqrt{3})^3} = \frac{\sqrt{6}}{3\sqrt{3}} = \frac{\sqrt{2}}{3}$$

Q4

a.

normalize $\vec{u}$: $\frac{\vec{u}}{\|\vec{u}\|} = \frac{(1,1)}{\sqrt{2}} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\nabla f = (-y\sin x,\ \cos x+2y)$

when $(x,y)=(0,1)$, $\nabla f_{(0,1)} = (0,3)$

$D_{\vec{u}} f = \boxed{\nabla f_{(0,1)}\cdot\frac{\vec{u}}{\|\vec{u}\|}} = (0,3)\cdot\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$

$= \frac{3}{\sqrt{2}}$

memorize this formula, which is a dot product, the output is a number

b

consider $F = z - e^z + 2xy - 3 = 0$

$\nabla F = (2y,\ 2x,\ 1-e^z)$

$(x,y,z) = (1,2,0)$, then $\nabla F = (4,2,0)$

tangent plane: need to memorize

$\boxed{\begin{array}{l} A(x-x_0)+B(y-y_0)+C(z-z_0)=0 \\ \text{passing point } (x_0,y_0,z_0) \\ \text{normal vector } (A,B,C) \end{array}}$

$4(x-1) + 2(y-2) + 0(z-0) = 0$

$4(x-1) + 2(y-2) = 0$

c

$F = x^2+2y^2+3z^2-498 = 0$

$\nabla F = (2x,\ 4y,\ 6z) = k(1,3,5)$

since tangent planes $\parallel$ $x+3y+5z-7=0$

$\begin{cases} x = \frac{1}{2}k \\ y = \frac{3}{4}k \\ z = \frac{5}{6}k \end{cases}$

so $F = (\frac{1}{2}k)^2 + 2(\frac{3}{4}k)^2 + 3(\frac{5}{6}k)^2 - 498 = 0$

$k = \pm 12$ $(k^2=144)$

passing tangent point $\pm(6,9,10)$

tangent plane:

$1(x+6)+3(y+9)+5(z+10)=0$ or

$1(x-6)+3(y-9)+5(z-10)=0$

or: $x+3y+5z\pm 83=0$

Q5

a

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{2-e^{xy}}-1}$$

$$= \lim_{(x,y)\to(0,0)} \frac{xy(\sqrt{2-e^{xy}}+1)}{(\sqrt{2-e^{xy}}-1)(\sqrt{2-e^{xy}}+1)}$$

$$= \lim_{(x,y)\to(0,0)} \frac{xy}{1-e^{xy}}\cdot(\sqrt{2-e^{xy}}+1)$$

set $a=xy$, $(x,y)\to(0,0)$ then $a\to 0$

$$= \lim_{a\to 0} \boxed{-\frac{a}{e^a-1}}\cdot(\sqrt{2-e^a}+1)$$

$$= -1\cdot(\sqrt{2-1}+1) = -1(1+1) = -2$$

note: $\lim_{a\to 0}\frac{a}{e^a-1} = 1$

b

set $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = \lim_{r\to 0^+} \frac{r^3(\cos^3\theta-\sin^3\theta)}{r^2}$$

$$= \lim_{r\to 0^+} r\underline{(\cos^3\theta-\sin^3\theta)} = 0$$

range $[-2,2]$

c

use two differant path derive two differant limit.

① consider $y=x$

$$\lim_{(x,y)\to(0,0)} \frac{xy^3}{x^2+y^6} = \lim_{x\to 0}\frac{x^4}{x^2+x^6} = \lim_{x\to 0}\frac{x^2}{1+x^4} = \frac{0}{1+0} = 0$$

② consider $x=y^3$

$$\lim_{(x,y)\to(0,0)} \frac{xy^3}{x^2+y^6} = \lim_{x\to 0}\frac{x^2}{x^2+x^2} = \lim_{x\to 0}\frac{x^2}{2x^2} = \frac{1}{2}$$

$0 \neq \frac{1}{2}$

lim does not exist.

Q6

(i)

$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = \lim_{r\to 0^+}\frac{r^2\cos\theta\sin\theta}{r} = \lim_{r\to 0^+} r\cos\theta\sin\theta = 0 = f(0,0)$$

(ii)

$$f_x = \lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0}\frac{\frac{h\cdot 0}{\sqrt{h^2+0}}-0}{h} = 0$$

$$f_y = \lim_{h\to 0}\frac{\frac{0\cdot h}{\sqrt{0+h^2}}-0}{h} = 0$$

Q7 see the solution

Q8

$y = mx^a$, $a<6$

approaching $(0,0)$:

① $m=0$, then $y=0$, $f=0$

② $m\neq 0$, if $y\leq 0$, $f=0$

③ $m\neq 0$, if $y>0$, then

$\frac{y}{x^6} = mx^{a-6} \longrightarrow +\infty$ as $x\to 0$

$\lim_{x\to 0} mx^{a-6} = \lim_{x\to 0} m\cdot\boxed{\frac{1}{x^{6-a}}} = +\infty$

$\to\infty$

so $y > x^6$, $f=0$

consider path $y = \frac{x^6}{2}$ s.t. $0<y<x^6$ $(x\neq 0)$

$\lim_{x\to 0} f(x,\frac{x^6}{2}) = 1 \neq 0 = f(0,0)$

so f is discontinuous at $(0,0)$

Q9.

Solution 9

Define $F(x,y,z,\lambda) = xy+2z-\lambda(x^2+y^2+z^2-11)$. Then $\nabla F = \vec{0}$ gives us

$$\begin{cases} y = 2\lambda x & (1) \\ x = 2\lambda y & (2) \\ 2 = 2\lambda z & (3) \\ x^2+y^2+z^2 = 11. & (4) \end{cases}$$

From (1) and (2): $x = 4\lambda^2 x \Rightarrow \underline{x(1-4\lambda^2)=0}$. $x=0$ or $1-4\lambda^2=0$

It is easy to forget case $x=0$

Case 1: $x=0$

Then (1) gives $y=0$. Constraint: $z^2=11 \Rightarrow z=\pm\sqrt{11}$.

Critical points: $(0,0,\sqrt{11})$, $(0,0,-\sqrt{11})$.

The corresponding values are $f=2\sqrt{11}$, $f=-2\sqrt{11}$.

Case 2: $\lambda = \pm\frac{1}{2}$

From (3): $z=\pm 2$. From (1) and (2): $y=\pm x$.

When $\lambda=\frac{1}{2}$, $z=2$, $y=x$, we have from (4): $2x^2+4=11 \Rightarrow x^2=\frac{7}{2}$.

Critical points: $\left(\sqrt{7/2},\sqrt{7/2},2\right)$, $\left(-\sqrt{7/2},-\sqrt{7/2},2\right)$.

The corresponding values are $f=\frac{7}{2}+4=\frac{15}{2}$.

When $\lambda=-\frac{1}{2}$, $z=-2$, $y=-x$, we have from (4): $2x^2+4=11 \Rightarrow x^2=\frac{7}{2}$.

Critical points: $\left(\sqrt{7/2},-\sqrt{7/2},-2\right)$, $\left(-\sqrt{7/2},\sqrt{7/2},-2\right)$.

The corresponding values are $f=-\frac{7}{2}-4=-\frac{15}{2}$.

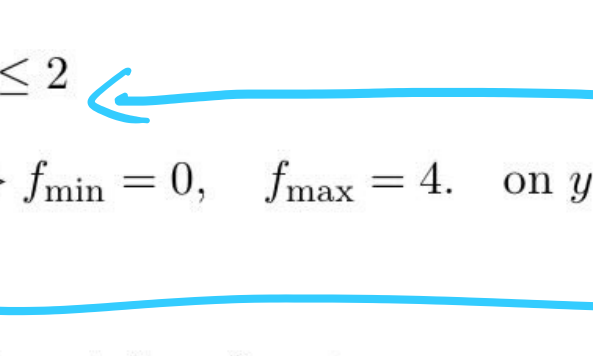
Comparing the above values, we have the following answer:

$$\boxed{f_{\max}=\frac{15}{2}}, \quad \boxed{f_{\min}=-\frac{15}{2}}.$$

Q10

Solution 10

Max/min of $f(x,y)=x^2+2y^2-x^2y^2$ in $D=\{x^2+y^2\leq 4,\ y\geq 0\}$.



① f_x or $f_y=0$

② boundary

(1)

$$\begin{cases} f_x = 2x-2xy^2=0 \\ f_y = 4y-2x^2y=0 \end{cases} \Rightarrow \begin{cases} x=\pm\sqrt{2} \\ y=1 \end{cases} \quad \text{and} \quad f(\pm\sqrt{2},1)=2.$$

These 2 points are inside D.

(2) on $y=0$, $f(x,0)=x^2$, $-2\leq x\leq 2$

$\Rightarrow f_{\min}=0,\quad f_{\max}=4.\quad$ on $y=0$.

(3) on $x^2+y^2=4$, $y>0$.

let $L(x,y,\lambda)=x^2+2y^2-x^2y^2+\lambda(x^2+y^2-4)$

$$\begin{cases} L_x = 2x-2xy^2+2\lambda x=0 \\ L_y = 4y-2x^2y+2\lambda y=0 \\ L_\lambda = x^2+y^2-4=0 \end{cases}$$

$$\begin{cases} 2x^2-2x^2y^2+2\lambda x^2=0 \\ 4y^2-2x^2y^2+2\lambda y^2=0 \\ x^2+y^2=4 \end{cases} \Rightarrow \begin{cases} x=\pm\sqrt{5/2} \\ y=\sqrt{3/2} \end{cases} \quad \text{or} \quad \begin{cases} x=0 \\ y=2 \end{cases}$$

$$f(\pm\sqrt{5/2},\sqrt{3/2})=\frac{7}{4}, \qquad f(0,2)=8.$$

(4) Compare all values $\Rightarrow$

$$\boxed{f_{\max}=8, \qquad f_{\min}=0}.$$